Where Are Your Parents? Exploring Potential Bias in Administrative Records on Children

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Abstract

This paper examines potential bias in the Census Household Composition Key's (CHCK) probabilistic parent-child linkages. By linking CHCK data to the American Community Survey (ACS), we reveal disparities in parent-child linkages among specific demographic groups and find that characteristics of children that can and cannot be linked to the CHCK vary considerably from the larger population. In particular, we find that children from low-income, less educated households and of Hispanic origin are less likely to be linked to a mother or a father in the CHCK. We also highlight some data considerations when using the CHCK.

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Introduction

Researchers rely heavily on survey data to analyze family structure; however, this approach rarely captures complex family structures and changes in these families over time. Large survey datasets, such as the Current Population Survey (CPS) or the American Community Survey (ACS), are cross-sectional and typically only allow for a static examination of households. Longitudinal surveys often capture rich data on family dynamics, but their relatively small sample sizes create issues of statistical precision especially when examining heterogeneous groups.[2] For example, the Fragile Families and Child Wellbeing Study collected data on births and families from 1998-2017, but only followed approximately 4800 children (Reichman et al, 2001). Even larger surveys, such as the Survey of Income and Program Participation (SIPP), with sample size ranging from 14,000 to 52,000 interviewed households, typically only capture a short window of data on families. In the case of the SIPP, households are followed for 2.5 to 4 years.

To bridge this gap in data coverage, it is possible to augment survey data with administrative records to better measure family relationships and to capture the experiences of individuals and family members across multiple points in time.[3] However, linking survey data to administrative data may limit analyses because it often fails to capture parents[4] and children who do not reside together. This could occur for many reasons: a parent is in the military, incarcerated, or homeless; a child lives with a parent only part-time or not at all due to custody arrangements; or a child is living with grandparents or other family members. Parents may also live apart due to the constraints of school and the labor market (Levin and Trost 1999). People who are tenuously attached to the residence are also likely to be omitted from the survey roster and altogether skipped by the standard survey process (Martin, 2007).

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[2] Several other surveys used in family dynamics research also have relatively small samples, such as the National Survey of Children, the National Survey of Families and Households, the National Longitudinal Survey of Youth, and the National Survey of America's Families.

[3] O'Hara, et al, (2017) outline several potential uses for linked data, including family members' labor force experiences (IRS data), health insurance coverage (Medicare and Medicaid data), and interactions with the criminal justice system (corrections data), as well as outcomes for veterans and military families (Veterans Affairs data). They also provide a thorough review of the literature that uses these linkages.

[4] We use the term "parent" to refer to individuals who have been listed as a parent on an application for a new Social Security Number (SSN) for a child. For biological children, this process typically occurs at birth. For adopted children, parents are encouraged to apply for new SSNs prior to or after completing the adoption. For more information, see Genadek et al, (2021).

A potential administrative data source to address the issue of non-residential parents is the Census Household Composition Key (CHCK). The CHCK is a unique Census Bureau data set that records nearly all births in the U.S. since 1997 with linkages to parents anonymously via the Census Bureau's assigned Protected Identification Key (PIK) (see Genadek et al, 2021 for a detailed description of the CHCK). The PIKs allow linkage to other datasets at the person level for both children and parents, without restrictions on first observing them together in a survey product or other administrative data source.

While the CHCK is an excellent tool for linking children to parents, it does not come without limitations. The Census Bureau creates the CHCK from annual extracts of the Social Security Administration's (SSAs) Numerical Identification (Numident) file beginning with the 2016 extract. The SSA Numident extract only contains the name of the parent(s) and does not provide the parent's Social Security Number (SSN). The Census Bureau creates the parental linkage in the CHCK by probabilistically linking children to their parents using the information on parent names from the Numident and address information contained in reference files built from a composite of trusted government administrative records.[5]

Using parental name and sex alone in the linkage process does not provide a great degree of certainty in the linkage (Luque & Wagner, 2015). To better identify a 'true' match to a mother or father, the process for linking parents is restricted to individuals who have shared an address with the child in administrative records reference files and decennial census data. The CHCK files use three vintages of address files: one that is typically the most current or corresponds to the CHCK year vintage, and two older files (typically 3-6 years older than the current vintage). The 2021 CHCK was produced before the 2020 decennial was processed, so only 2010 decennial data is included in the creation process for CHCK vintages up to 2021. However, matching on first and last name using string-matching algorithms – even while blocking on address – can be challenging for various ethnicities with complex naming structures or unique naming systems (Treeratpituk & Giles, 2012). This may introduce statistical bias into analyses using only children with linkable parents.

[5] These reference files are referred to as the GeoBase reference files, which combine information from the Social Security Administration's Numident file with information on addresses from other administrative records, including IRS 1040s and 1099s. For more information, see Mulrow et al (2011) or Wagner and Layne (2014).

Luque & Wagner (2015) assessed the coverage and quality of parental linkages in a prototype Census Kidlink Database that was produced in 2007. They found that younger children and noncitizens had a higher percentage of cases where neither parent could be linked to the child. They also found that when a mother or father's PIKs were missing in the Kidlink Database, the mothers' and fathers' names were missing for only 0.3% and 17.5% of these records, respectively. A missing name indicates the parent was not listed on the child's birth certificate. The discrepancy between fathers and mothers is likely related to increasing rates of births to unmarried women who choose not to list the father on the birth certificate. In 2021, 40 percent of all births were to nonmarried women (Osterman et al, 2023). However, missing names only accounts for a minority of PIK missingness. Thus, PIK missingness was not caused primarily by lack of name availability, so PIKs were not assigned to parents for other reasons such as lack of a name match or lack of co-residence with the child. Current research using the CHCK linked to ACS data has shown that of children with a PIK in the ACS, approximately 95% are in the CHCK and linked to a mother in the CHCK, but a smaller proportion (80-85%) reside with the mother they are linked to in the CHCK in the ACS (Genadek et al, 2022)[6].

The purpose of this paper is to explore the possibility and nature of systemic bias that could be introduced from the probabilistic process of linking parents to children in the CHCK. With the CHCK data linked to the ACS, the rich survey data allow us to examine how linkage rates differ by detailed demographic characteristics and produce new estimates on which factors are correlated with parental PIK missingness in the CHCK.. We find disparities in parent-child linkages among specific socio-demographic groups and find that ACS reported characteristics of children that can and cannot be linked to the CHCK vary considerably from the larger population. In particular, we find that children from low-income, less educated households and of Hispanic origin are less likely to be linked to a mother or a father in the CHCK.

[6] This indicates children are experiencing changes in co-residence between birth and when they are observed in the ACS. Future research will examine the fluidity of children's living arrangements.

Data

Special Considerations when using the CHCK

After their first appearance in the CHCK, a small number of children link to new parents in subsequent CHCK files.[7] This can occur when a parent is later added to a birth certificate, a child is adopted, or other information is updated with the SSA. Changes also happen due to co-residential data improving over time, sometimes referred to as the "tax code effect" (Luque and Wagner, 2015). Most administrative records in a given year come from tax data, and since children born in the current year are typically not claimed until the next tax filing year, there can be a lag in the availability of this data. Previous research has also noted that young children are less likely to have linkages to parents in the CHCK (Luque & Wagner, 2015; Genadek et al, 2022). For this paper, we use the most recent information as that is most likely to be reflected in the survey data, however researchers may want to use the first linkage available, for example to most accurately link children to birth parents when studying fertility (Genadek et al, 2022).

There were also substantial updates to the source data for the CHCK starting in 2018 (Genadek et al, 2022). These stem from improvements in production of the Numident at SSA in preparation for the 2020 Census. For this reason, it is recommended to avoid using the 2016 and 2017 vintages for any children that can be found in later vintages of CHCK files. It should also be noted that the CHCK files include the variable names "Mother PIK" and "Father PIK." Census receives information from the SSA with these columns labeled in this fashion, so prior to 2021 the process for parental linkages in the CHCK required that these labeled column match on the reported sex (female or male) for mothers and fathers. However, working closely with SSA, the Census Bureau discovered that even though these fields are labeled in this manner, they may not necessarily reference mother and/or father. Same sex parents can appear in either field. For this reason, the Census Bureau's production record linkage team removed the sex restriction on these fields starting with the 2021 CHCK vintage.[8]

For the 2016 to 2021 CHCK files, the parental PIKs are left blank if the linkage process fails to determine a match. However, this does not mean that there is not a parent listed in the

[7] For our sample of children, 1% have a change in Mother PIK and 0.7% have a change in Father PIK during the time they are observed in the CHCK.

[8] Our analysis of the practical implications of this sex restriction is the subject of a follow-up paper. For this analysis, we match mother and/or father PIK fields in the CHCK to corresponding biological mother or father by sex.

Numident extract; it simply means there was no match found. Starting in 2022, the CHCK includes a flag for how complete the name information was in the raw Numident data for either parent.[9] This is valuable for researchers who wish to understand if parents are present in SSA records but are not necessarily linked through the probabilistic process.

Children may fail to link to a parent at higher rates through several channels, introducing linkage bias. Previous research suggests that certain characteristics are associated with failure to assign a PIK: being a recent immigrant, being highly mobile with frequent address changes, being of Hispanic origin, or not paying taxes due to being low income or working under the table (Bond et al, 2014). This is also true if identifying information (such as first and last name) differs between the survey and reference files. This fact is of particular importance to the CHCK linkage, since the only identifying information we have on the parents is first and last names.

As noted earlier, matching on first and last name using string-matching algorithms can be challenging for various ethnicities with complex naming structures or unique naming systems. This is especially true for Hispanic, Arabic, and East Asian populations (Treeratpituk & Giles, 2012). Table 1 gives examples of how individuals may choose to represent their given names. If the linkage process fails to determine a match based on parental names and co-residency, the PIK for the parent is left blank.

Table 1: Example of different types of name variations for different ethnicities

Name	Variation
Fatimah bint Tariq bin Khalid al-Fulan	Fatimah al-Fulan
Pedro Juan Lopez Rodrıguez	Pedro Lopez
Mao Zedong	Mao Tse-tung
Heung-Yeung Shum	Harry Shum
Li Wei Gang	Weigang Li

Source: Treeratpituk and Giles (2012).

Sampling Children and Households

The CHCK has no demographic information beyond age, so we build our sample of children from the American Community Survey (ACS). The ACS is an annual sample of about

[9] These flags consist of a string of three digits ranging from "000" (first, middle and last name fields on the Numident were blank) to "111" (all name fields were non-blank). Various combinations of zeros and ones indicate the missingness from the Numident. For example, if all name fields except middle name were non-blank, the flag is assigned a value of "101."

3.5 million households and is one of the most comprehensive sources of information about the U.S. population, including a wide variety of social, demographic, and housing information. As the largest federal household survey, the ACS is important for understanding populations at the state or sub-state level. From the ACS, we observe information about parents and children on race, household income, geography, education, and place of birth. For our analysis, we use person and housing unit data from the 2014-2018 1-year ACS.

We limit the sample to households with at least one child under the age of 18 at the time of the survey. We drop any children within a household who are assigned the same PIK and children under the age of 18 who are listed as a spouse, an unmarried partner, or householder. Although later years of the 1-year ACS are available, we specifically end our sample in 2018 to reduce the number of very young children that are not yet captured in administrative reference files.[10] For this reason, we end our ACS sample in 2018 but use the CHCK files up to 2022 to better capture information on young children.

The CHCK is produced yearly starting in 2016, however we only use the CHCK for 2018 onward. We stack each of the 2018 to 2022 yearly CHCK files and overwrite any changes in parental information with the newest version if it differs from the previous year. This gives us a near universe of all children in the Numident born between 1999 and 2021 which we subsequently link to our sample of children drawn from the ACS using PIKs. For the full sample of over 5.3 million children from the five years of ACS data, 91.9% of them have a PIK and of those, 93.8% appear in the CHCK.[11]

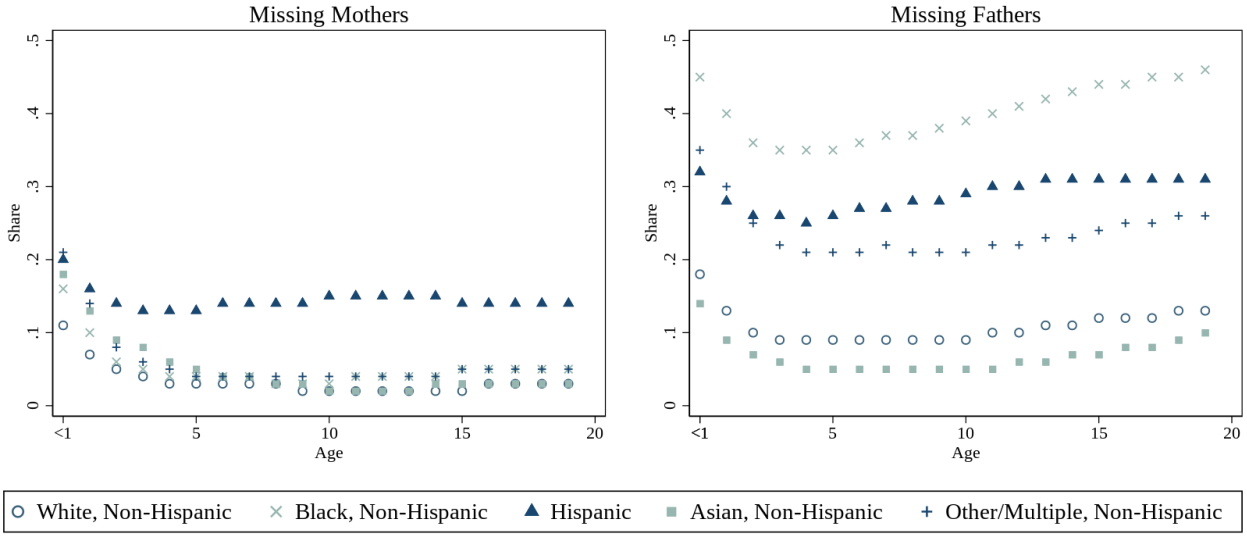
[10] See discussion in the prior section.

[11] The majority of these cases in which a child has a PIK and is not present in the CHCK are when a child is born in or earlier than 1999 and excluded due to age restrictions on the CHCK.

Linkage Assessment and Analysis

We first note some basic linkage statistics. Our sample of children from the 2014-2018 1-year ACS that links to the 2018-2022 CHCK contains 4,570,000 children. Of these, 3.8% are missing a mother's PIK and 14.6% are missing a father's PIK on the CHCK.[12] For mothers, we expect their CHCK PIK missingness to be solely due to failed probabilistic linkage or mothers who do not appear in administrative data (no SSN or ITIN). For fathers with missing CHCK PIKs, this can be a combination of factors: the father was not on the birth certificate entirely, they do not appear in administrative data, or there was a failed probabilistic linkage.[13]

Figure 1: Share Missing CHCK Parent by Child's Age and Child's Race



Source: 2014-2018 1-year American Community Survey, 2018-2022 Census Household Composition Key.

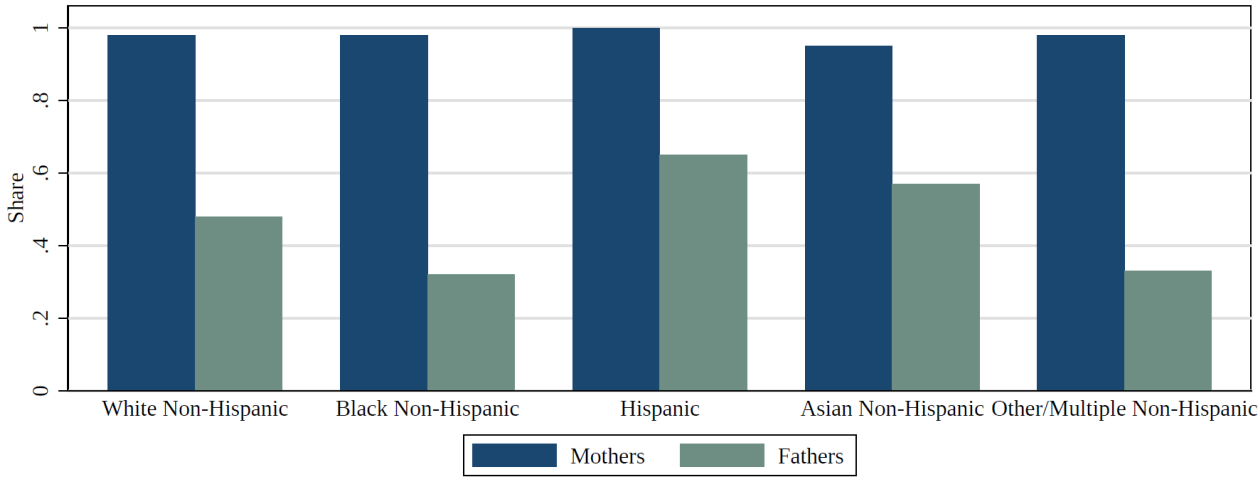
Since age and race are of special consideration, Figure 1 shows how the linkage to either a mother or father varies over the age and race of the child. For this figure, we use age of the child from stacked CHCK vintages linked to the race of the child given in the ACS. Overall, mothers are missing at lower rates than fathers, although this varies by the child's race. Linkage rates improve for both parents over the first four years of a child's life. Except for very young children (<4 years of age), mothers of Hispanic children are missing at rates almost double of

[12] While we use the terms "mother's PIK" and "father's PIK" in this instance, it is important to note that these are not sex specific for cases of same-sex parents.
[13] Failed probabilistic linkages can be the result of different factors, including differences in naming conventions, unstable addresses, or other factors (see Bond, et al 2014).

other races, while fathers of Black children are missing at the highest rates over all other races/ethnicities.

For those children who appear in the 2022 CHCK vintage, we observe how many had parents that could not be linked but had some parental name information given in the SSA Numident. For those missing mothers, 99% of them had at least a partial parental name listed while 49.8% of those with missing fathers had some name information. However, this varies by race/ethnicity of the child. Figure 2 highlights these differences by child's race. Even when a parental PIK is missing, there is some name data for at least 95% of mothers. For fathers with missing PIKs, this ranges across the race of the child: 32% of black children have non-missing father name information and 67% of Hispanic children.

Figure 2: Share of Parents with Non-missing Name Information
for Children Missing a CHCK Parent PIK by Child's Race/Ethnicity



Source: 2014-2018 American Community Survey, 2022 Census Household Composition Key. Figure reports the subset of children with missing linkages to a mother or father in the CHCK.

When determining the nature of bias in CHCK linkages, we are especially interested when children are missing a mother. We expect to find children linked to both parents or a mother alone since the majority of children are born to married couples or unmarried women (Smock and Schwartz, 2020). Unmarried women may or may not choose to list a father on the birth certificate. On the other hand, mothers are present at birth and generally listed on the birth certificate, so we can assume that those missing a mother – especially considering the results from Figure 2 – are the result of some bias in the linkage discussed in the previous section.

Incomplete or inaccurate linkages in the CHCK become more apparent when we look at the child's relationship to the reference person in the ACS. By using the ACS relationship variable, we can directly observe children with a reported biological parent. We then limit the sample to children with a definite link to an ACS biological mother or father[14] (reference person parent and a child's relationship reported as biological). We then link the children in the ACS to the CHCK when they were assigned a PIK. We compare the PIKs in the ACS for the biological parent to the parent's PIK in the CHCK for the sample child. For the ACS biological mothers, 93.2% have the same PIK in both the ACS and CHCK. However, we find that 2.7% of mothers have conflicting PIKs between the CHCK and the ACS, 1.4% have a PIK listed in the ACS alone, 1.7% have a PIK in the CHCK alone, and 1% have no PIK in either source. For biological fathers, we find 91% have the same PIK in both sources, while 2.6% have different PIKs, 3.6% have PIKs in the ACS alone, 1.4% have a CHCK PIK alone, and 1.1% have no PIK in either source.

Table 2. Share in PIK Status Category
ACS Parents Linked to Biological Children

	Same PIK	Different PIKs	ACS PIK only	CHCK PIK only	No PIK	Total	N
Biological Mother	93.2	2.7	1.4	1.7	1.0	100	2,020,000
Biological Father	91.2	2.6	3.6	1.4	1.1	100	1,790,000

Sources: 2014-2018 1-year ACS, 2018-2022 CHCK. Unweighted sample includes children in the ACS under 18 who have a PIK and are linked to the CHCK. All children in this sample are linked as a biological child to the ACS reference person.

To better understand the source of these mismatches, we disaggregate biological parents by race in Table 3. We find that both Hispanic mothers and fathers have higher proportions in which PIKs differ between CHCK and ACS, those that have a PIK in only one data source, and those that have no PIK at all. In particular, 6.0% of Hispanic mothers and 6.5% of Hispanic fathers are missing PIKs in both sources. The proportions in this category are less than 1% for all other races. We also find that of Hispanic fathers, 6.0% have PIK mismatches and 10.5% have PIKs on the ACS alone.

[14] For Tables 2 and 3, we assume that the sex of the reference person (biological parent) matches the sex column of the CHCK and link female reference persons with "Mother PIK" and male reference persons with "Father PIK." The ACS does some imputation of both reference person and child relationships. When we perform the analysis and exclude any biological parents and children whose relationship has been imputed, the results are substantially the same.

These statistics should be taken into consideration when linking children to parents. The survey data provides direct measures of parenthood, though these are self-reported and are expected to have some error. In addition, it may be the case that the PIKs assigned to parents in the survey data, such as the ACS, have less error because they are assigned using more information on parents to base the PIK assignment on. Researchers may want to prioritize the parent and child PIK pairings in survey products rather than the child-parent PIK pairings obtained from the CHCK alone.

Table 3. Share in PIK Status Category by Parents' Race
ACS Parents Linked to Biological Children

	Same PIK	Different PIKs	ACS PIK only	CHCK PIK only	No PIK	Total	N
Biological Mother's Race							
White Non-Hispanic	96.2	2.0	0.7	1.0	0.1	100	1,348,000
Black Non-Hispanic	94.3	2.9	1.0	1.7	0.1	100	23,1000
Hispanic	80.3	4.7	4.8	4.2	6.0	100	32,7000
Asian Non-Hispanic	93.3	3.2	1.0	2.2	0.2	100	76,000
Other/Multiple Non-Hispanic	90.6	4.3	1.2	3.5	0.4	100	37,500
Biological Father's Race							
White Non-Hispanic	95.2	1.8	2.0	0.9	0.1	100	1,265,000
Black Non-Hispanic	86.6	3.7	7.6	1.6	0.5	100	95,500
Hispanic	73.4	6.0	10.5	3.5	6.5	100	270,000
Asian Non-Hispanic	94.5	2.0	1.6	1.7	0.2	100	136,000
Other/Multiple Non-Hispanic	85.3	5.3	5.6	3.1	0.7	100	24,000

Sources: 2014-2018 1-year ACS, 2018-2022 CHCK. Unweighted sample includes children in the ACS under 18 who have a PIK and are linked to the CHCK. All children in this sample are linked as a biological child to the ACS reference person. Rows may not sum to 100 due to rounding.

Probability of non-linkage to a CHCK parent

Up to this point, we have just described summary statistics for children and parents and their various linkages. Given the information from the previous section, we examine how the likelihood of not observing a parent in the CHCK data varies with demographics and socio-economic status of children and their household. Children's household composition may have changed between when they were observed in the CHCK, at the time of birth, and when they are in the ACS survey due to parental separation or co-residence with a nonparent relative.

To assess how different factors are associated with the likelihood of linkage to a CHCK parent, we estimate regression models separately for mothers and fathers. Our dependent variable

is equal to one if a child is linked to a mother (or father) in the CHCK. We estimate a linear probability model, where the covariates are a series of categorical variables that include:

1. Child's demographic characteristics: sex, race, and whether they were foreign-born
2. Socio-economic status of the household: quintile of household income and highest level of educational attainment within the household
3. Geographic characteristics: Census Division and urban status

Race categories include white non-Hispanic, black non-Hispanic, Hispanic, Asian non-Hispanic, and Other/Multiple non-Hispanic. Household income quintiles are constructed using the total household income (adjusted for inflation). Quintiles are computed within state to allow for regional differences. We also include indicators for each of the nine Census Divisions as well as an indicator for living in an urban area.

Our results are found in Table 4. For mothers, we expect missingness to be solely due to failed probabilistic linkage or mothers who do not appear in administrative data (no SSN or ITIN). For missing fathers, this can be a combination of factors: the father was not on the birth certificate entirely, they do not appear in administrative data, or there was a failed probabilistic linkage. [15] For the following discussion, we present results that are unweighted. In Table 4, we also include results obtained using ACS weights that have been adjusted for PIK missingness by inverse probability weighting (IPW).[16] Although including weights does not affect the sign or significance, the magnitudes of the coefficients are slightly different than the unweighted estimates.

We find no significant correlation on the sex of the child when children are missing a mother, however female children are 0.5 percentage points (pp) more likely to be missing a father than male children. Compared to white children, Black non-Hispanic children are 0.3pp *less* likely to be missing a mother, but 22.1pp *more* likely to be missing a father. We also find that Hispanic children are 7.9 and 11.7pp more likely to be missing a mother or father, respectively. For other/multiple non-Hispanic races, children were 0.2pp more likely to be missing a mother and 6.9pp more likely to be missing a father than white non-Hispanic children.

[15] Failed probabilistic linkages can be the result of different factors, including differences in naming conventions, unstable addresses, or other factors (see Bond, et al 2014).

[16] We use inverse probability weighting with a logistic regression to model the probability of each observation's presence or absence base on observed covariates. These probabilities are then inverted to assign weights, which are applied to the dataset to adjust the ACS provided weights to mitigate bias due to missing data. In this instance, children without PIKs assigned are not included in the analysis.

Asian non-Hispanic children were less likely to be missing a father (1.5pp lower) and the effects for mothers were higher, though small (0.1pp). This is likely tied to trends in nonmarital births. In 2021, 70 percent of births to Black non-Hispanic women and 53 percent of births to Hispanic women were to nonmarried women compared to 26 percent of white women (Osterman et al. 2023). Children who are foreign born are more likely to be missing a mother (2.1pp) and be missing a father (0.3pp).

Household characteristics are also correlated with parental missingness in the CHCK. We find that for both mothers and fathers, missingness decreases as the child lives in households farther up the income distribution. For example, children in the top quintile of the income distribution are 1.8pp less likely to be missing mothers and 12.6pp less likely to be missing fathers than those in the bottom of the income distribution. We also see the same pattern for household educational attainment. Children are less likely to be missing parental linkages as the maximum educational attainment of the household increases. When the household's maximum educational attainment was a high school degree or GED, children were 10.0pp and 7.2pp less likely to be missing a mother or father linkage, respectively, over those households who had no high school completion. This jumps to 12.8pp less likely for mothers and 15.6pp less likely for fathers for households with a bachelor's degree.

Linkage missingness also varies over geography and urban/rural. Although there is no effect for mothers, children in urban areas are 1.1 pp more likely to be missing a father. For geography, our comparison is the New England region. All other regions are more likely to have missing mothers and fathers, although this varies in magnitude. For fathers, children in households who live in the East South Central region (4.2pp), West South Central (2.3pp), West North Central (1.9pp), and South Atlantic (1.9pp) have the highest comparative rates of missingness. Children living in the Pacific, the East South Central, and South Atlantic regions were 0.9pp, 0.8pp, and 0.7pp more likely to have missing mothers than those living in New England.

Discussion

Researchers often link administrative sources to survey data to study family structure, but these datasets often leave out children who do not reside with their parents. The CHCK addresses this issue by recording most births in the United States and linking those births to parents via

PIK. However, while the CHCK fills an important role in capturing children who do not reside with their parents, we find the CHCK is not representative of the national population because of bias in linkable records.

Researchers using the CHCK in analysis should keep in mind that the characteristics of children that can and cannot be linked to the CHCK vary considerably from the larger population. Because linkages improve over the first four years of a child's life, researchers may want to select different vintages of the CHCK based on their population of interest. Fathers are also more likely to be missing than mothers. In particular, we find that children from low-income, less educated households and of Hispanic origin are less likely to be linked to a mother or a father in the CHCK. For most children in the CHCK without a link to a mother, we find the mother has some name information on the child's Numident record, but a PIK was not assigned. This is likely related to challenges with name matching due to complex naming structures as described in Table 1.

Choosing a weighting scheme for final analysis to account for linkage bias is also important. We showed in the simple regression analysis case that IPW used to correct for PIK missingness can alter the resulting coefficients. However, different weighting approaches are required depending on the missing data pattern and mechanism. In using the CHCK linked to children – and thus to parents – in the ACS, we encounter two levels of missingness. First, a child may be missing a PIK due to their characteristics. The second level of missingness occurs at the parent level based on parents' characteristics. While the statistical implications of the second level of missingness is beyond the purview of this paper, it is worth further consideration. When using a sample that consists of only those children linked to parents in the CHCK, additional weighting might be required.

With these limitations in mind, the CHCK can still be a useful administrative source of parent-child linkages for use in studying how family structure and household characteristics change over time. Researchers should consider using adjusted survey weights and controlling for demographic characteristics to alleviate bias when conducting analyses.

Table 4. Likelihood of Missing CHCK Linkage

	Missing Mother				Missing Father			
	Unweighted		Weighted		Unweighted		Weighted	
Child's Sex	Mean	Probability	Mean	Probability	Mean	Probability	Mean	Probability
Male	0.038		0.050		0.143		0.175	
Female		-0.000**		0.000		0.005***		0.006***
		(0.000)		(0.000)		(0.000)		(0.000)
Child's Race								
White Non-Hispanic	0.017		0.018		0.086		0.093	
Black Non-Hispanic		-0.003***		-0.008***		0.221***		0.211***
		(0.000)		(0.000)		(0.001)		(0.001)
Hispanic		0.079***		0.091***		0.117***		0.122***
		(0.000)		(0.000)		(0.000)		(0.001)
Asian Non-Hispanic		0.001**		0.001*		-0.015***		-0.016***
		(0.000)		(0.000)		(0.001)		(0.001)
Other/Multiple Non-Hispanic		0.002**		0.011***		0.069***		0.067***
		(0.001)		(0.001)		(0.001)		(0.002)
Child is Foreign Born								
No	0.037		0.048		0.146		0.177	
Yes		0.021***		0.022***		0.003***		-0.006***
		(0.001)		(0.001)		(0.001)		(0.001)
Quintile of HH income distribution								
Bottom Quintile	0.073		0.092		0.298		0.336	
Second Quintile		-0.003***		-0.003***		-0.069***		-0.070***
		(0.000)		(0.001)		(0.001)		(0.001)
Third Quintile		-0.013***		-0.017***		-0.102***		-0.110***
		(0.000)		(0.001)		(0.001)		(0.001)
Fourth Quintile		-0.017***		-0.021***		-0.119***		-0.130***
		(0.000)		(0.000)		(0.001)		(0.001)
Top Quintile		-0.018***		-0.021***		-0.126***		-0.137***
		(0.000)		(0.000)		(0.001)		(0.001)
Maximum educational attainment within HH								
No HS or GED	0.190		0.223		0.366		0.409	
HS degree/GED		-0.100***		-0.111***		-0.072***		-0.082***
		(0.000)		(0.001)		(0.001)		(0.001)
Some college - no degree		-0.125***		-0.145***		-0.099***		-0.115***
		(0.000)		(0.001)		(0.001)		(0.001)
Associates degree		-0.130***		-0.151***		-0.122***		-0.136***
		(0.000)		(0.001)		(0.001)		(0.002)
Bachelor's degree		-0.128***		-0.146***		-0.156***		-0.173***
		(0.000)		(0.001)		(0.001)		(0.001)
Graduate or Professional degree		-0.127***		-0.145***		-0.167***		-0.186***
		(0.000)		(0.001)		(0.001)		(0.001)
Census Region								
New England	0.025		0.031		0.103		0.126	
Middle Atlantic		0.001**		0.003***		0.011***		0.014***
		(0.000)		(0.001)		(0.001)		(0.001)
East North Central		0.001*		0.004***		0.013***		0.019***
		(0.000)		(0.001)		(0.001)		(0.001)
West North Central		0.005***		0.008***		0.019***		0.026***
		(0.001)		(0.001)		(0.001)		(0.001)
South Atlantic		0.007***		0.012***		0.019***		0.023***
		(0.000)		(0.001)		(0.001)		(0.001)
East South Central		0.008***		0.012***		0.042***		0.048***

		(0.001)		(0.001)		(0.001)		(0.001)
West South Central		0.002***		0.002**		0.023***		0.020***
		(0.000)		(0.001)		(0.001)		(0.001)
Mountain		0.005***		0.006***		0.007***		0.009***
		(0.001)		(0.001)		(0.001)		(0.001)
Pacific		0.009***		0.012***		0.000		0.001
		(0.000)		(0.001)		(0.001)		(0.001)
Urban Status								
Rural	0.028		0.034		0.121		0.137	
Urban		0.000		0.000		0.011***		0.012***
		(0.000)		(0.000)		(0.000)		(0.001)
Constant		0.146***		0.163***		0.283***		0.305***
		(0.001)		(0.001)		(0.001)		(0.002)

Sources: 2014-2018 1-year ACS, 2018-2022 CHCK.

N=4572000. Sample includes children in the ACS under 18 who have a PIK and are present in the CHCK.

Standard errors in parentheses. * p<0.05, ** p<0.01, and *** p<0.001

Weighted results use initial ACS weights and adjust for PIK missingness via inverse probability weighting.

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